

Mathematical analysis 3

Chapter 3 : Special Functions

Gamma and Beta functions

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Course outline

1 Gamma function

2 Beta function

Definition of Gamma function

In this chapter, we introduce the classical Gamma function, essentially understood as a generalized factorial. There are many applications of this function, for example, it is found in analysis, number theory, probability, and fractional calculus.

Definition

Let Γ be the function defined on $]0, +\infty[$ by:

$$\Gamma(x) = \int_0^{+\infty} t^{x-1} e^{-t} dt.$$

Example:

$$\Gamma(1) = \int_0^{+\infty} t^{1-1} e^{-t} dt = \int_0^{+\infty} e^{-t} dt = -e^{-t} \Big|_0^{+\infty} = 1.$$

Properties of Gamma function

We have the following properties:

- 1 Γ is a defined and positive function on $]0, +\infty[$.
- 2 Γ is continuous on $]0, +\infty[$.
- 3 Γ is differentiable on $]0, +\infty[$ and

$$\forall x \in]0, +\infty[: \Gamma'(x) = \int_0^{+\infty} \ln t \cdot t^{x-1} e^{-t} dt.$$

Properties of Gamma function

Proof: For all $t \in]0, +\infty[$:

$$t^{x-1} = e^{\ln(t^{x-1})} = e^{(x-1)\ln(t)}$$

By setting $\phi(x, t) = t^{x-1} e^{-t}$, we obtain:

$$\begin{aligned} \Gamma'(x) &= \frac{d\Gamma}{dx}(x) = \int_0^{+\infty} \frac{\partial \phi(x, t)}{\partial x} dt = \int_0^{+\infty} \ln(t) e^{(x-1)\ln(t)} e^{-t} dt \\ &= \int_0^{+\infty} \ln(t) t^{x-1} e^{-t} dt \end{aligned}$$

- ④ Γ is of class C^∞ on $]0, +\infty[$ and for all $x \in]0, +\infty[$,

$$\forall k \in \mathbb{N} : \Gamma^{(k)}(x) = \int_0^{+\infty} (\ln(t))^k t^{x-1} e^{-t} dt.$$

- ⑤ Γ recurrence relation (the functional equation of Γ):

$$\forall x \in]0, +\infty[, \Gamma(x+1) = x\Gamma(x)$$

Properties of Gamma function

Proof: Integrating Γ by parts:

$$u(t) = e^{-t}, \quad v'(t) = t^{x-1} \Rightarrow u'(t) = -e^{-t}, \quad v(t) = \frac{1}{x} t^x,$$

for all $x \in]0, +\infty[$, we obtain

$$\begin{aligned} \Gamma(x) &= \lim_{a \rightarrow +\infty} \left(\frac{1}{x} t^x e^{-t} \Big|_0^a \right) + \int_0^{+\infty} \frac{1}{x} t^x e^{-t} dt \\ &= \lim_{a \rightarrow +\infty} \left(\frac{1}{x} a^x e^{-a} \right) + \frac{1}{x} \int_0^{+\infty} t^x e^{-t} dt \\ &= \frac{1}{x} \Gamma(x+1). \end{aligned}$$

Where:

$$\lim_{a \rightarrow +\infty} \frac{1}{x} a^x e^{-a} = \lim_{a \rightarrow +\infty} \frac{1}{x} e^{x \ln a - a} = \lim_{a \rightarrow +\infty} \frac{1}{x} e^{a(x \frac{\ln a}{a} - 1)} = 0.$$

Properties of Gamma function

- ⑥ The function Γ is the extension of the factorial function to $x \in]0, +\infty[$. For all $n \in \mathbb{N}$: $\Gamma(n+1) = n!$.

Proof: For $n > 0$, we have

$$\Gamma(n+1) = n\Gamma(n) = n(n-1)\Gamma(n-1)$$

$$= n(n-1)(n-2)\Gamma(n-2) = \dots = n(n-1)(n-2)\dots 3 \times 2 \times \Gamma(2) = n!.$$

Since we agree that $0! = 1 = \Gamma(1) = \Gamma(0+1)$, we deduce

$$\forall n \in \mathbb{N} : \Gamma(n+1) = n!,$$

and we can write

$$\forall n \in \mathbb{N} \setminus \{0\} : \Gamma(n) = (n-1)!.$$

Properties of Gamma function

7 We have $\Gamma\left(\frac{1}{2}\right) = \sqrt{\pi}$.

Proof: $\Gamma\left(\frac{1}{2}\right) = \int_0^{\infty} t^{\frac{1}{2}-1} e^{-t} dt = \int_0^{\infty} t^{-\frac{1}{2}} e^{-t} dt.$

Let $t^{\frac{1}{2}} = y \Rightarrow t = y^2$ and $dt = 2ydy$, thus

$$\Gamma\left(\frac{1}{2}\right) = \int_0^{\infty} y^{-1} e^{-\frac{y^2}{2}} 2ydy = 2 \int_0^{\infty} e^{-\frac{y^2}{2}} dy = 2 \left(\frac{\sqrt{\pi}}{2} \right) = \sqrt{\pi}.$$

Properties of Gamma function

- 8 For all $k \in \mathbb{N}$: $\Gamma\left(k + \frac{1}{2}\right) = \frac{(2k)!}{2^{2k} \cdot k!} \sqrt{\pi}$.

Proof: Knowing that $\Gamma(x+1) = x\Gamma(x)$, for all $k \in \mathbb{N}$, we obtain:

$$\begin{aligned}
 \Gamma\left(k + \frac{1}{2}\right) &= \left(k - \frac{1}{2}\right) \cdot \Gamma\left(k - \frac{1}{2}\right) \\
 &= \left(k - \frac{1}{2}\right) \left(k - \frac{3}{2}\right) \Gamma\left(k - \frac{3}{2}\right) \\
 &\vdots \\
 &= \left(k - \frac{1}{2}\right) \left(k - \frac{3}{2}\right) \cdots \frac{3}{2} \cdot \frac{1}{2} \cdot \Gamma\left(\frac{1}{2}\right) \\
 &= \left(k - \frac{1}{2}\right) \left(k - \frac{3}{2}\right) \cdots \frac{3}{2} \cdot \frac{1}{2} \cdot \sqrt{\pi} \\
 &= \left(\frac{2k-1}{2}\right) \left(\frac{2k-3}{2}\right) \cdots \frac{3}{2} \cdot \frac{1}{2} \cdot \sqrt{\pi}
 \end{aligned}$$

On the other hand, we have

$$k! = k(k-1)(k-2) \times \dots \times 3 \times 2 \times 1 = \frac{2k}{2} \cdot \frac{2k-2}{2} \cdot \frac{2k-4}{2} \cdot \dots \cdot \frac{4}{2} \cdot \frac{2}{2},$$

so

$$\Gamma\left(k + \frac{1}{2}\right) = \left(\frac{2k}{2} \cdot \frac{2k-1}{2} \cdot \frac{2k-2}{2} \cdot \dots \cdot \frac{3}{2} \cdot \frac{2}{2} \cdot \frac{1}{2}\right) \cdot \frac{\sqrt{\pi}}{k!}$$

therefore

$$\Gamma\left(k + \frac{1}{2}\right) = \frac{(2k)!}{2^{2k} \cdot k!} \sqrt{\pi}.$$

The negative case

The negative case

- ① Thanks to the recurrence relation $\Gamma(x+1) = x\Gamma(x)$, we can set by convention

$$\forall x \in]-1, 0[: \Gamma(x) = \frac{\Gamma(x+1)}{x}.$$

Example:

$$\Gamma\left(-\frac{1}{2}\right) = \frac{\Gamma\left(\frac{1}{2}\right)}{\left(-\frac{1}{2}\right)} = -2\sqrt{\pi}.$$

- ② Let $n \in \mathbb{N}$, for any non-integer negative value of x , such that $x \in]-n, -n+1[$, we have

$$\Gamma(x) = \frac{\Gamma(x+n)}{x(x+1)(x+2)\dots(x+n-1)},$$

and $\Gamma(x)$ has the sign of $(-1)^n$.

- ③ For all $n \in \mathbb{N}$: $\Gamma\left(-n + \frac{1}{2}\right) = (-1)^n \frac{2^{2n} \cdot n!}{(2n)!} \sqrt{\pi}.$

Course outline

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2 Beta function

Definition of Beta function

Definition

For all $x, y \in]0, +\infty[$, we define the function Beta by:

$$\beta(x, y) = \int_0^1 t^{x-1} (1-t)^{y-1} dt.$$

Example:

- ① $\beta(1, 1) = \int_0^1 t^{1-1} (1-t)^{1-1} dt = \int_0^1 1 dt = 1.$
- ② $\beta(2, 1) = \int_0^1 t^{2-1} (1-t)^{1-1} dt = \int_0^1 t dt = \frac{1}{2}.$
- ③ $\beta(1, 2) = \int_0^1 t^{1-1} (1-t)^{2-1} dt = \int_0^1 (1-t) dt = \frac{1}{2}.$
- ④ Show that $\beta(2, 2) = \frac{1}{6}.$

Definition of Beta function

Example: Calculate $\beta\left(\frac{1}{2}, \frac{1}{2}\right)$.

Solution: We have

$$\beta\left(\frac{1}{2}, \frac{1}{2}\right) = \int_0^1 t^{-\frac{1}{2}} (1-t)^{-\frac{1}{2}} dt,$$

using the change $t^{\frac{1}{2}} = x \Rightarrow t = x^2$ and $dt = 2x dx$, then

$$\begin{aligned}\beta\left(\frac{1}{2}, \frac{1}{2}\right) &= \int_0^1 t^{-\frac{1}{2}} (1-t)^{-\frac{1}{2}} dt = \int_0^1 x^{-1} (1-x^2)^{-\frac{1}{2}} 2x dx \\ &= 2 \int_0^1 \frac{1}{\sqrt{1-x^2}} dx = 2[\arcsin x]_0^1 = 2\left(\frac{\pi}{2}\right) = \pi.\end{aligned}$$

Proposition

We have

$$\forall x, y \in]0, +\infty[: \beta(x, y) = \beta(y, x)$$

Proof: We have

$$\beta(x, y) = \int_0^1 t^{x-1} (1-t)^{y-1} dt,$$

let's set $s = 1 - t$, then we obtain $t = 1 - s$, $dt = -ds$, $t = 0 \Rightarrow s = 1$ and $t = 1 \Rightarrow s = 0$, this gives us

$$\begin{aligned} \beta(x, y) &= \int_0^1 t^{x-1} (1-t)^{y-1} dt = - \int_1^0 (1-s)^{x-1} s^{y-1} ds \\ &= \int_0^1 s^{y-1} (1-s)^{x-1} ds = \beta(y, x). \end{aligned}$$

Relation between the functions Γ and β

Proposition

For all $x, y \in]0, +\infty[$:

$$\beta(x, y) = \frac{\Gamma(x)\Gamma(y)}{\Gamma(x+y)}. \quad (2.13)$$

Proof: Consider for x, y strictly positive:

$$\begin{aligned} \Gamma(x)\Gamma(y) &= \int_0^{+\infty} t^{x-1} e^{-t} dt \int_0^{+\infty} s^{y-1} e^{-s} ds \\ &= \int_0^{+\infty} t^{x-1} e^{-t} dt \int_0^{+\infty} s^{y-1} e^{-s} ds \\ &= \int_0^{+\infty} \int_0^{+\infty} t^{x-1} s^{y-1} e^{-(t+s)} dt ds \\ &= \iint_D t^{x-1} s^{y-1} e^{-(t+s)} ds dt, \end{aligned}$$

where $D = \{(t, s) : t > 0, \text{ and } s > 0\}$.

Let's use the following change of variables:

$$\begin{aligned} u &= t + s \\ v &= \frac{t}{t + s}, \end{aligned}$$

which implies $t = u \cdot v$ and $s = u - u \cdot v$. We deduce that $\Delta = \{(u, v) : u > 0, \text{ and } 0 < v < 1\}$ and the Jacobian

$$J = \begin{vmatrix} \frac{\partial t}{\partial u} & \frac{\partial t}{\partial v} \\ \frac{\partial s}{\partial u} & \frac{\partial s}{\partial v} \end{vmatrix} = \begin{vmatrix} v & u \\ 1 - v & -u \end{vmatrix} = -u$$

So, $dt ds = |J| du dv = u du dv$, which gives

$$\begin{aligned} \Gamma(x)\Gamma(y) &= \iint_{\Delta} u^{x+y-1} v^{x-1} (1-v)^{y-1} e^{-u} u du dv \\ &= \left(\int_0^{+\infty} u^{x+y-1} e^{-u} du \right) \left(\int_0^1 v^{x-1} (1-v)^{y-1} dv \right) \\ &= \Gamma(x+y) \times \beta(x, y). \end{aligned}$$

Proposition

We have

$$\textcircled{1} \quad \forall x, y > 0 : \beta(x, y) = \int_0^{+\infty} \frac{s^{x-1}}{(1+s)^{x+y}} ds.$$

$$\textcircled{2} \quad \forall x, y > 0 : \beta(x, y) = \int_0^1 \frac{s^{x-1} + s^{y-1}}{(1+s)^{x+y}} ds.$$

$$\textcircled{3} \quad \beta(x, y) = 2 \int_0^{\frac{\pi}{2}} (\sin \theta)^{2x-1} (\cos \theta)^{2y-1} d\theta.$$

$$\textcircled{4} \quad \forall x, y > 0 : \beta(x, y+1) = \frac{y}{x+y} \beta(x, y), \text{ and } \beta(x+1, y) = \frac{x}{x+y} \beta(x, y).$$

$$\textcircled{5} \quad \beta(x, y) \cdot \beta(x+y, 1-y) = \frac{\pi}{x \sin(\pi y)}.$$

Proposition

For $0 < x < 1$, we have:

$$\beta(x, 1-x) = \frac{\pi}{\sin(\pi x)}.$$

Proposition (Stirling formula)

For $x > 0$ we have

$$x! \sim x^x e^{-x} \sqrt{2\pi x}, \quad (x \rightarrow +\infty).$$

In other words

$$\Gamma(x+1) \sim x^x e^{-x} \sqrt{2\pi x}, \quad (x \rightarrow +\infty).$$

We have the following limits:

$$\lim_{x \rightarrow +\infty} \Gamma(x) = +\infty$$

$$\lim_{x \rightarrow +\infty} \frac{\Gamma(x)}{x} = +\infty$$

$$\lim_{x \rightarrow 0} \Gamma(x) = +\infty$$